Quantitative Analysis of Neocortical Gyrencephaly in African Elephants (*Loxodonta africana*) and Six Species of Cetaceans: Comparison With Other Mammals

Paul R. Manger,[1]* Michelle Prowse,[1] Mark Haagensen,[2] and Jason Hemingway[1]

[1]School of Anatomical Sciences, Faculty of Health Sciences, University of the Witwatersrand, Parktown, Johannesburg, Republic of South Africa

[2]Department of Radiology, University of the Witwatersrand–Donald Gordon Medical Centre, Parktown, Johannesburg, Republic of South Africa

ABSTRACT

This study provides quantitative data on the extent of gyrencephaly in the large-brained African elephant and several species of cetaceans (from smaller to larger brained) in comparison with other mammals. Across three mammalian orders (primates, carnivores, and artiodactyls), the species with the larger brains are more gyrencephalic with each order, exhibiting a specific negative allometry. The African elephant, with a 5-kg brain, has a gyrencephalic index (GI) of 3.89, which, though highly gyrencephalic, is not more so than would be predicted for a mammal with a 5-kg brain. The cetaceans had an average GI of 5.43, are the most gyrencephalic mammals studied to date, and are more gyrencephalic than one would predict based on comparison with other mammals. No relationship between brain mass and GI was evident in the cetaceans as seen in other mammals, with all cetaceans showing similar GIs irrespective of brain mass (range of GI 5.23–5.70, range of brain mass 577–5617 g). This is yet another parameter indicating cetaceans to be neuroanatomical outliers. Two species of pinnipeds studied had GIs that were well above those seen for terrestrial carnivores, and the aquatic manatee was close to lissencephalic. Thus, all three groups of marine mammals showed unusual extents of cortical gyrencephaly, indicating a morphological alteration of the telencephalon associated with the return to the marine environment. The analysis suggests that cortical thickness and neuronal density are important factors in determining the extent of gyrencephaly across mammalian species. J. Comp. Neurol. 520:2430–2439, 2012.

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The vast majority of mammalian brains are small, with a lissencephalic cerebral cortex. In spite of this, there are several mammalian groups with larger brains, such as primates, carnivores, and artiodactyls, that exhibit varying degrees of gyrencephaly. The examination of gyral and sulcal morphology and patterns has a long history; however, a comprehensive review of the literature indicates that, whereas larger brains appear to be more gyrencephalic, there are order-specific features in the pattern of gyrification (Welker, 1990). Zilles et al. (1989) provided the first comprehensive quantitative analysis of gyrencephaly and, through the determination of a gyrencephalic index (GI), demonstrated that in primates larger brains are indeed more gyrencephalic than smaller brains (with the presence of a negative allometry). A more recent study (Pillay and Manger, 2007) confirmed the findings of Zilles et al. (1989) for primates and extended this by demonstrating that there are order-specific patterns to the quantifiable parameters of gyrencephaly such that each mammalian order examined demonstrated a unique and

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*CORRESPONDENCE TO: Paul R. Manger, School of Anatomical Sciences, Faculty of Health Sciences, University of the Witwatersrand, 7 York Road, Parktown, 2193 Johannesburg, Republic of South Africa. E-mail: paul.manger@wits.ac.za

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specific negative allometry of GI associated with brain mass. What is clear, despite the order specificity (Manger, 2005), is that, in all mammalian orders, generally the species with the largest brains have the greatest gyrencephaly. To date, however, those mammals with the largest brains, elephants and cetaceans, have not been examined quantitatively in this respect.

Qualitative impressions of the surface, and sections through the cerebral cortex, of elephant and cetacean brains indicate that the cerebral hemispheres of these species are highly gyrencephalic (see, e.g., Cozzi et al., 2001; Shoshani et al., 2006; Oelschläger and Oelschläger, 2009). The aim of the current study is to fill the gaps in our knowledge of the quantitative GI measurements for elephants and cetaceans and to identify how the relationship between GI and brain mass in cetaceans and elephants compares with that of species from other mammalian orders. In addition to the elephants and cetaceans examined, the current study includes data from other Afrotherian (hyrax and manatee) and aquatic mammals (pinnipeds and manatee) in order to determine whether some causal factors in gyrencephaly might emerge. Any potential causal factors that emerge from this broader comparative approach may help our understanding of why and how much the cortex develops fissures and folds.

MATERIALS AND METHODS

Specimens

Data from 91 individual mammals, encompassing 72 species, were used in this study (Table 1). Data were obtained for three adult male African elephants (Manger et al., 2009); a rock hyrax; a West Indian manatee; six species of cetaceans; four species of artiodactyl, 15 species of carnivores, including two pinniped species; and 44 species of primates. Permission to obtain the elephant specimens, targeted as part of a planned and authorized cull, was granted by the Zimbabwe Parks and Wildlife Management Authority. Ethical permission was obtained from the University of the Witwatersrand Animal Ethics Committee, and the animals were treated and used according to the guidelines of this committee, which parallel those set down by the NIH for use of animals in scientific experiments. Permission was granted by the Malilangwe Nature Conservation Trust to obtain African elephants resident on their property near Chiredzi in southeastern Zimbabwe. All work with the elephants was performed under the direction and supervision of an extensively experienced wildlife veterinarian employed by the Malilangwe Trust. The data for the artiodactyls, five species of carnivores, and primates were taken from the literature (Zilles et al., 1989; Pillay and Manger, 2007), whereas those for the Afrotherians; cetaceans; and 10 carnivores, including two pinnipeds, were generated specifically for the current study (see below). The data generated for the African elephants were derived from magnetic resonance images (MRI; see below), whereas the data generated for the hyrax, manatee, bottlenose dolphin (40 equally spaced sections from rostral to caudal poles of the cerebral hemisphere), and additional carnivores were derived from images downloaded from the website of the Comparative Mammalian Brain Collections of the University of Wisconsin and Michigan State University (http://www.brainmuseum.org). Images of the cerebral cortex of the humpback whale used to calculate the gyrencephalic index were taken from Hof and van der Gucht (2007; five equally spaced parasagittal sections from medial to lateral in the hemisphere), for the common dolphin from Oelschläger et al. (2008; five equally spaced sections from the rostral diencephalon to the caudal pole of cerebral hemisphere), for the dwarf sperm whale from Oelschläger et al. (2010; six equally spaced sections from the rostral diencephalon to the caudal pole of cerebral hemisphere), for the beluga whale from Marino et al. (2001; six equally spaced sections from rostral to caudal poles of the cerebral hemisphere), and for the killer whale from Marino et al. (2004; 10 equally spaced sections from rostral to caudal poles of the cerebral hemisphere). Obtaining a full representation of the cerebral hemispheres for quantification in the cetacean brains was not always readily available, so three of our data points come from partial, though well-represented, portions of the cerebral hemispheres (for the humpback whale, common dolphin, and dwarf sperm whale). It is known that the extent of gyrencephaly differs across the cerebral hemisphere (Armstrong et al., 1991; Rilling and Insel, 1999), but the consistency in the data obtained for the brains of the cetaceans indicates that the data generated herein for these species in which only partial hemispheres were analyzed are likely to be an accurate reflection of the entire hemisphere. Brain mass was taken from these studies for each of the species when provided, but when not provided the average species brain mass was used (Manger, 2006).

MRI

Brains from three African elephants (*Loxodonta africana*, LA1, LA2, LA3; Manger et al., 2009) underwent MRI to obtain images from which the gyrencephalic index could be calculated. The brains were fixed and stored as described by Manger et al. (2009) and then scanned in coronal, sagittal, and horizontal planes. The specimens were scanned on a Philips 1.5 Tesla Intera System (Philips, Eindhoven, The Netherlands), using all three elements of the head and neck coil. The brains were

TABLE 1.

List of the Species, Brain Mass, and Gyrification Indices Analyzed in the Current Study

Scientific name	Common name	Brain mass (g)	GI	Source[1]
Afrotherians				
Loxodonta africana	African elephant (LA1)	5,145	4.01	1
Loxodonta africana	African elephant (LA2)	5,250	3.85	1
Loxodonta africana	African elephant (LA3)	4,835	3.81	1
Procavia capensis	Rock hyrax	16	1.38	1
Trichechus manatus	West Indian manatee	350	1.07	1
Cetaceans				
Kogia sima	Dwarf sperm whale	577	5.26	1
Delphinus delphis	Common dolphin	981	5.41	1
Tursiops truncatus	Bottlenose dolphin	1,530	5.63	1
Delphinapterus leucas	Beluga whale	1,871	5.23	1
Orcinus orca	Killer whale	5,617	5.70	1
Megaptera novaeangliae	Humpback whale	4,600	5.35	1
Artiodactyls				
Sus scrofa	Domestic pig	95.3	2.16	2
Odocoileus virginianus	White-tailed deer	160	2.27	2
Lama glama	Llama	200.3	2.70	2
Bos taurus indicus	Zebu	474	2.53	2
Carnivores				
Mustela erminea	Ermine	4.00	1.33	1
Mustela putorius	European polecat	8.3	1.36	1
Neovison vison	American mink	8.5	1.46	1
Cynictis penicillata	Meerkat	14.53	1.35	2
Bassariscus astutus	Ringtail	20.70	1.46	1
Galictis vittata	Greater grisson	24.30	1.59	1
Felis catus	Domestic cat	36.9	1.50	2
Nasua narica	White-nosed coati	37.00	1.62	1
Ailurus fulgens	Lesser panda	41.68	1.51	1
Vulpes vulpes	Red fox	43.50	1.66	1
Crocuta crocuta	Hyena	162.5	1.74	2
Panthera leo	African lion	258	1.85	2
Ursus maritimus	Polar bear	458.6	2.04	2
Pinnipeds				
Phoca vitulina	Harbor seal	275	2.38	1
Zalophus californianus	California sea lion	379.1	2.52	1
Primates				
Sixty-one individuals	Forty-four species	1.7–1,400	1.1–2.99	2,3

[1]Sources for gyrencephalic index: 1) this study; 2) Pillay and Manger (2007); 3) Zilles et al. (1989).

removed from their storage containers, drained of excess fluid, and placed in the head coil wrapped in a dry sheet, thus being exposed directly to air, which also partially entered the ventricles. After different scan parameters had been tested, the following sequence was selected as giving the best detail and the least artefact (especially at air–fluid interfaces). The selected T1-weighted inversion recovery sequence consisting of 2 mm slices without a gap, had a TR (time to repeat) of between 6.5 and 10.9 seconds depending on the number of slices, a time to echo (TE) of 10 msec, and a time to invert (TI) of 300 msec. The number of signal averages varied between three and four, with a flip angle of 90° and an echo train of 10. The scan times varied between 15 and 25 minutes. The antifreeze liquid in which the brains were stored showed high signal on both T1- and T2-weighted sequences, and the routine clinical T1 and T2 sequences produced very similar T2-like images of the brain specimens. This is possibly related to the lack of water in the tissues of the specimen secondary to the fixation and storage process. The images were processed using the freely available open source software program OsiriX (Rosset et al., 2004; www.osirix-viewer.com), which allowed us to convert the DICOM-format images to JPEG images for subsequent analysis.

Calculation of GI

The GI is a quantitative calculation of the degree of folding of the cerebral cortex. For all data presented in the current study, the GI was calculated according to the method described by Zilles et al. (1989), which according to Pillay and Manger (2007) appears to be the most reliable and simplest method for cross-species comparisons. The GI, as described by Zilles et al. (1989), is a comparison of the complete (pial surface) and outer cortical contours of the cerebral neocortex (Fig. 1). The GI was

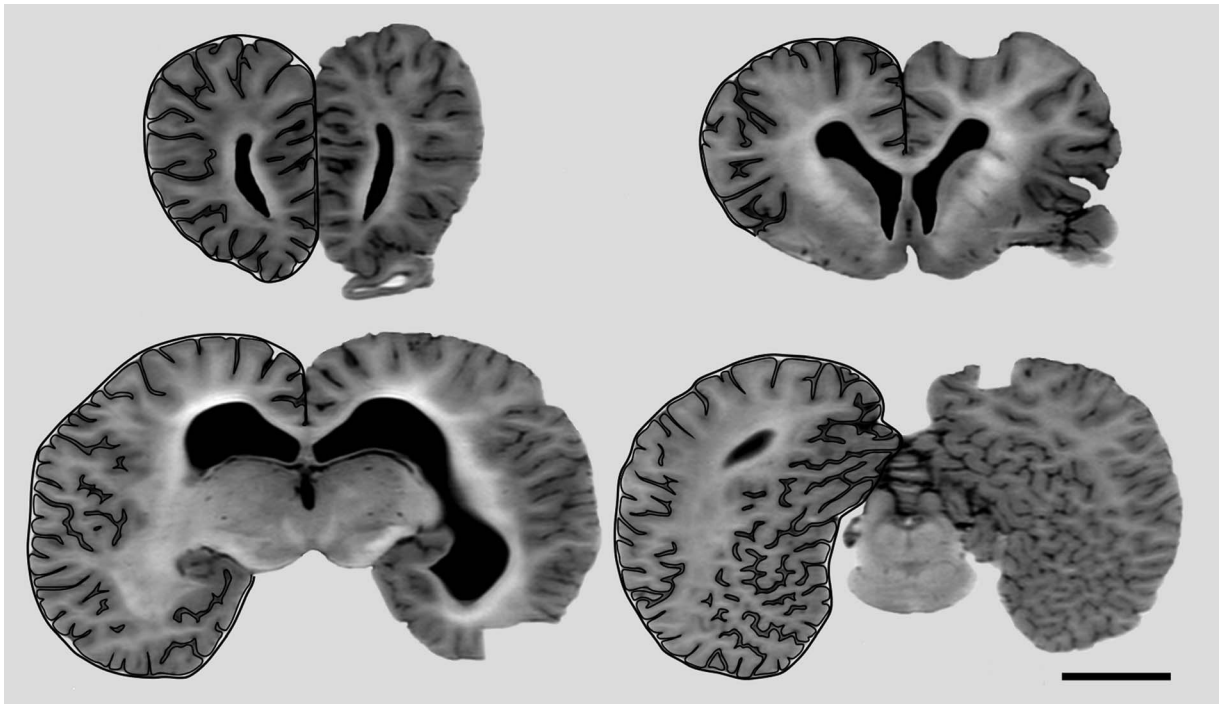


Figure 1. MR images showing the elephant cerebral cortex at different levels in the coronal plane. Examples of the outlines used to determine the outer and complete contours of the cerebral cortex are drawn on the left hemisphere of each image. Scale bar = 5 cm.

created by dividing the perimeter of the complete pial contour by the perimeter of the outer cortical contour. Depending on the size of the brain and availability of images, between 30 and 40 evenly spaced sections through an entire cerebral hemisphere were analyzed for each specimen (fewer images were available for the cetaceans, for which images were taken from published studies; see above for details regarding sampling for the cetaceans). The lengths of these contours were measured using the freely available Image J software, and the sum of the perimeters of the complete pial contour was divided by the sum of the outer contours.

Statistical analysis

The brain mass and GI data were log transformed, and Pearson's correlation coefficient was calculated for the primates and carnivores to determine the relationship between brain mass and GI. Least squares regressions were calculated to construct 95% prediction intervals with the aim of evaluating the probability of the elephants, cetaceans, pinnipeds, hyrax, and manatee being predicted from the calculated slopes for either primates or carnivores. In the current study, we opted not to undertake analyses that take into account phylogenetic nonindependence (such as independent contrasts) for two specific reasons. The first is that a phylogeny that is truly correct and encompasses the range of species studied does not appear to be available as yet. Phylogenetic data that may be incorrect, or indeed missing (in which branch lengths of one would have to be inserted into the data set), would introduce major errors into the analysis that we wished to avoid. To overcome this potential shortcoming, we have provided the entire database (Table 1) so that anyone wishing to undertake such an analysis has the raw data available. Second, we do find that phylogeny is involved in the extent of gyrencephaly, but it appear to be at the ordinal level rather that at the species level. Therefore, although independent contrast methods may strengthen the finding of ordinal variation in gyrencephaly, they will not alter the conclusions made in the current study. This is a more prudent course of action in terms of the analysis, and we feel that at the current time it is the most appropriate.

RESULTS

The results obtained for the relationship between GI and brain mass for the carnivores (excluding pinnipeds), artiodactyls, and primates are similar to those obtained in previous studies for these orders (Zilles et al., 1989; Pillay and Manger, 2007). The relationship between GI and brain mass for both the elephants and the hyrax appear to be what one would predict for mammals with their brain masses, whereas the GI of the manatee is significantly lower than that of other mammals with a similar

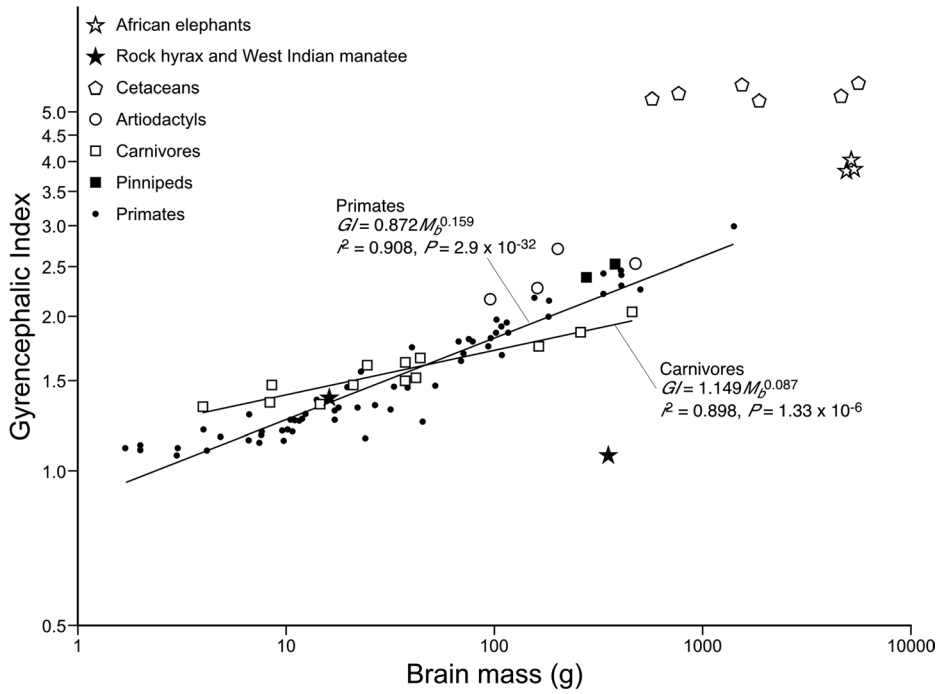


Figure 2. Graphic plot depicting the relationship between brain mass (grams) and gyrencephalic index (GI) in the species examined in the current study. The graph shows the raw data on log-transformed scales with the associated equations describing the regression lines generated for primates and carnivores. Mb, body mass; P, probability of the two variables being uncorrelated; r^2, coefficient of determination.

brain mass. The GI for the six species of cetaceans analyzed in the current study was found to be extremely high for a mammal with this brain mass, making their cortex the most gyrencephalic among the mammalian orders investigated. Finally, in comparison with that of other carnivores, the GI of both pinniped species was high for their brain mass.

Primates, carnivores, and artiodactyls

The current study used data from two published sources for the primates (Zilles et al., 1989; Pillay and Manger, 2007) and analyzed brain mass and GI for 44 species (61 individuals; Table 1). The analysis demonstrated a strong correlation between brain mass and GI for species from this order, this relationship having a negative allometry (slope = 0.159, $r^2 = 0.908$, $P = 2.9 \times 10^{-32}$; Fig. 2). The negative allometry predicts that, for every doubling in brain mass, primate GI will increase by approximately 1.11 times, with 90.8% of the variation in GI being accounted for by variation in brain mass. For the carnivores, we used previously published data (Pillay and Manger, 2007; five species) and added eight additional species (Table 1). The analysis demonstrated a strong correlation between brain mass and GI for carnivores, showing a negative allometry (slope = 0.087, $r^2 = 0.898$, $P = 1.33 \times 10^{-6}$; Fig. 2). The negative allometry predicts that, for every doubling in brain mass, carnivore GI will increase by approximately 1.06 times, with 89.8% of the variation in GI being accounted for by variation in brain mass. No statistically significant relationship could be found for the four artiodactyl species analyzed in the current study (data from Pillay and Manger, 2007), this lack of a relationship potentially resulting from the small number of data points (Fig. 2); however, it appears that a similar increase in GI is evident with increasing brain mass in the artiodactyls. The analysis of data from these three mammalian orders provided a solid baseline against which to compare data from Afrotherians, cetaceans, and pinnipeds.

African elephants

The quantitative measurement of the African elephant GI, averaging 3.89, was found to be what is expected for this species in comparison with other mammals based on their large brain mass (average = 5,076.66 g; Table 1, Fig. 2). The data for elephants fell within the 95% prediction intervals for primates (Fig. 3) but above that for carnivores with respect to their relationship between brain mass and GI. A qualitative impression of the data appears to indicate that the elephants have a GI similar to what would be expected for an artiodactyl with an approximately 5-kg brain (Figs. 2, 3). Thus, although the GI of the elephants is large, it is no larger than would be expected for either a primate or an artiodactyl with a large brain.

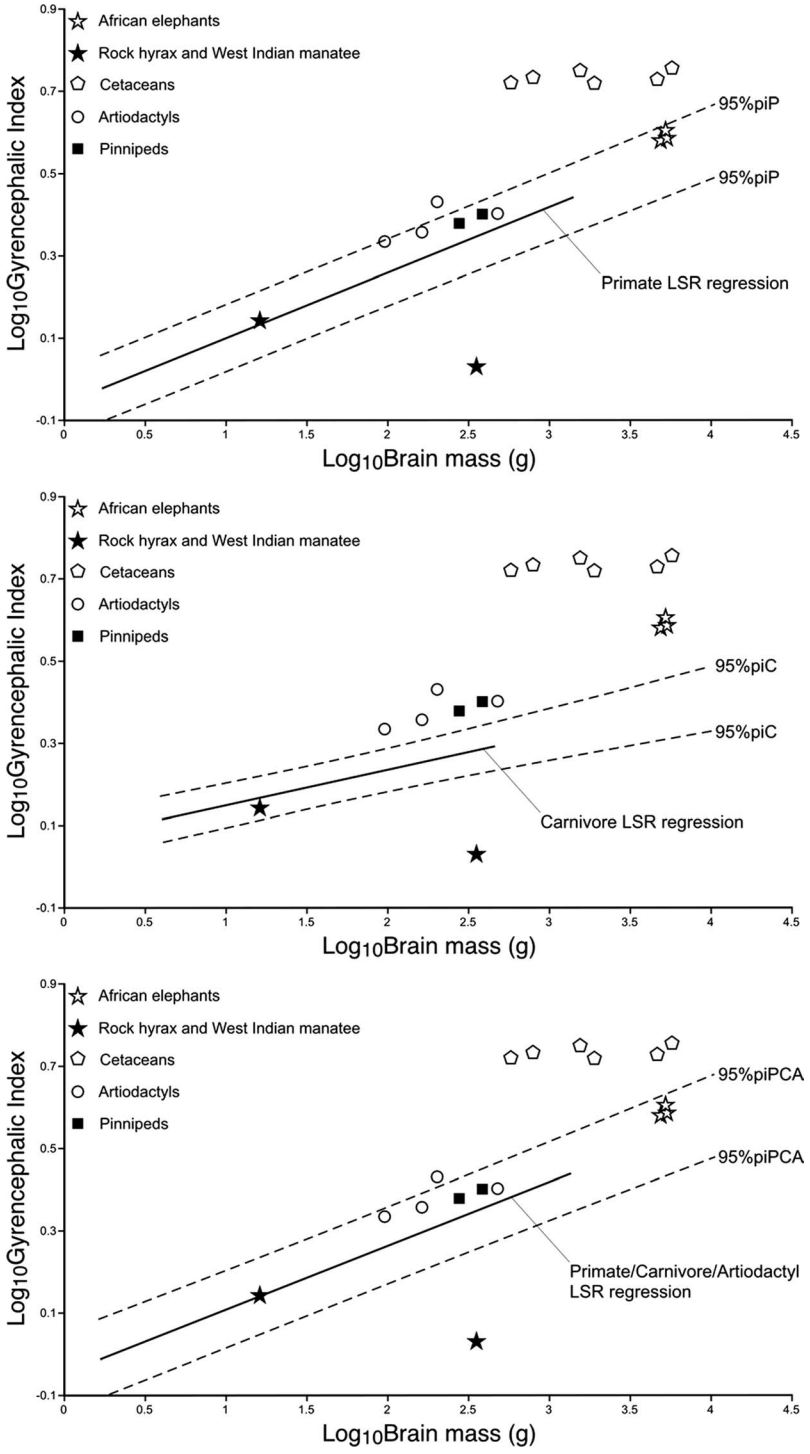


Figure 3. Graphic plots depicting the relationships between gyrencephalic index (GI) and brain mass on log-transformed data in the species investigated; 95% prediction intervals (PI) were plotted on the graphs based on the primate data (upper graph), carnivore data (middle graph), and a combination of primate + carnivore + artiodactyl data (lower graph). Note that, for both the primate and combination data, the GI of the elephants falls within the 95% PIs but above the 95% PI for carnivores. The hyrax data point falls within the 95% PIs of all three groups, and the manatee data point falls below the 95% PIs of all three groups, underscoring the lissencephalic nature of the manatee cerebral cortex. For cetaceans, the GIs are above the 95% PIs of primates, carnivores, and the combined data, being the most gyrencephalic mammals analyzed in the current study and underscoring their status as neuroanatomical outliers. Pinnipeds fall within the 95% PIs for primates and the combined data but above those of the closely related carnivores, emphasizing their unusual status within the carnivores. All three groups associated with the marine environment appear to have abnormal gyrencephaly of the cerebral cortex.

Rock hyrax and West Indian manatee

The GI of the rock hyrax, being 1.38, was observed to fall within the 95% prediction intervals for primates and carnivores, indicating that the extent of cortical gyrencephaly in this species is what one would expect for a mammal with a 16-g brain. The West Indian manatee, well know for its large (350 g) but lissencephalic brain (Welker, 1990), has a GI of 1.07 (correlating closely to the value of 1.06 provided by Reep and O'shea, 1990). In all cases, the data for this species fell below the 95% prediction intervals generated for all groups of mammals, emphasizing the lissencephalic nature of the cerebral cortex in this species.

Cetaceans

The six species of cetaceans analyzed in the current study were observed to have the most gyrencephalic cerebral cortices of all mammalian species studied to date. The GI for cetaceans averaged 5.43 (range 5.23–5.70, range of brain mass 577–5,617 g, almost an order of magnitude difference; Table 1, Fig. 2). In addition to this overall high GI for cetaceans, there was no relationship between brain mass and GI as seen in the other mammalian orders examined, rather the GI did not appear to increase with increasing brain mass. Thus, for cetaceans, both the absolute GI and the scaling of GI with brain mass (i.e., it does not change with changes in brain mass) are different from those of all other mammalian orders examined. The GI observed for the given brain mass in cetaceans was significantly different from that in primates and carnivores and fell well above the 95% prediction intervals for these two orders, and visual inspection of the data (Figs. 2, 3) shows the GI to be well above that of all other species, including those with similar brain masses.

Harbor seal and California sea lion

Gyrencephalic indices of 2.38 and 2.52 were calculated for the harbor seal and California sea lion, respectively (Table 1, Fig. 2), values that lie close to those of artiodactyls of similar brain masses. Although the data for these two pinnipeds fell within the 95% prediction intervals calculated for primates, they fall above the 95% prediction intervals calculated on the basis of other carnivore species (Fig. 3). Thus, although they are carnivores, the pinnipeds appears to have cerebral cortices that are significantly more gyrencephalic than other members of the same mammalian order.

DISCUSSION

The current study confirms previous findings (Zilles et al., 1989; Pillay and Manger, 2007) and reveals two novel findings of interest. The first novel finding is that the large brain of the African elephant, though highly gyrencephalic, is not more so than would be predicted for a mammal with an approximately 5-kg brain. The second finding of interest is that in all species studied that are associated with an aquatic environment (cetaceans, manatee, and pinnipeds), the calculated GIs are unusual in relation to mammals in general and/or to closely related mammalian species.

Gyrencephaly of mammals in general

The current study confirms and extends, through the addition of more data, previous findings related to the extent of gyrencephaly in mammals. Although qualitative observations have indicated that mammals with larger brains are more gyrencephalic than those with smaller brains (Welker, 1990), Zilles et al. (1989) were the first to quantify this parameter across a range of primate species. Zilles et al. (1989) demonstrated a negative allometry of the gyrencephalic index of primates when brain size increased, such that larger brains were more gyrencephalic than smaller brains (in the current study, we calculate that, for every doubling in primate brain mass, the extent of gyrencephaly increases 1.11 times). Moreover, Zilles et al. indicated that there does appear to be a difference in the scaling of GI and brain mass between prosimian and simian primates. Although this difference was not examined in the current study, it is a noteworthy finding of intraordinal variation for the primates. The current study and that of Pillay and Manger (2007) confirmed the observations made by Zilles et al. (1989) for primates and extended those findings into other mammalian orders. Rodents, carnivores, and artiodactyls also appear to follow the general assumption that larger mammalian brains are more gyrencephalic, but each mammalian order shows its own specific relationship between GI and brain mass (Pillay and Manger, 2007). Thus, when comparing gyrencephaly across mammalian species, there appear to be three general features of note: 1) larger mammalian brains are more gyrencephalic than smaller brains (Zilles et al., 1989; Pillay and Manger, 2007); 2) each mammalian order appears to have a unique scaling relating to the extent of gyrencephaly (Pillay and Manger, 2007); and 3) each mammalian order appears to present with its own unique pattern of gyral and sulcal formation (Welker, 1990). Thus, in terms of both qualitative and quantitative measures, mammals in general appear to show order specificity (Manger, 2005). These three points are generalized features of mammalian gyrencephaly, and the exceptions are discussed below.

Gyrencephaly of the African elephant neocortex

The GI of the elephant cerebral cortex was found to average 3.89, which in comparison with the other mammals

studied at first appears to be quite high. In spite of this, an analysis of the data shows that this value is what might be expected for a mammal with an approximately 5-kg brain. Though more gyrencephalic than would be expected for a carnivore with a 5-kg brain, the elephant is as gyrencephalic as would be expected for a primate with a 5-kg brain. This result confirms previous qualitative observations that the cerebral cortex of the elephant is significantly gyrencephalic (Cozzi et al., 2001; Shoshani et al., 2006) but tempers this conclusion with the demonstration that it is not more gyrencephalic than would be expected for a mammal of its brain size. There is, therefore, nothing specifically unusual about the extent of the gyrencephaly observed in the extant elephant brain. Similarly, with the measure of the GI of the rock hyrax, nothing specifically unusual for a mammal with a brain of approximately 16 g arose from the analysis.

Gyrencephaly of aquatic mammals

In all three instances of aquatic mammals examined in the current study, the GI was unusual compared with the GI of either other mammals (cetaceans and manatee, fully aquatic mammals) or closely related mammals (pinnipeds, semiaquatic carnivores). In two cases, cetaceans and pinnipeds, the GI is relatively higher, the cortex being more gyrencephalic than expected for a mammal of their brain mass. For the manatee, the GI is extremely low, a fact noted previously (Reep and O'shea, 1990). Cetaceans not only showed a very high GI but there was no correlation between brain mass and GI as seen in other mammals; rather, GI for cetaceans does not appear to change when there are changes in brain mass. In the present study, the smallest cetacean brain analyzed was almost 10 times smaller than the largest analyzed. This difference in scaling of GI and brain mass within cetaceans correlates with previous observations on the cerebral cortex, in which, instead of scaling in an allometric fashion as seen in other mammals (Finlay and Darlington, 1995), the surface area of cetacean cortex scales in an isometric fashion (Ridgway and Brownson, 1984). Thus, in terms of both absolute GI and the scaling of GI with brain mass, the cetaceans are neuroanatomical outliers, contrary to assertions made previously (Marino et al., 2004).

The two pinnipeds are clearly more gyrencephalic than the remainder of carnivores examined in the current study. The increase in gyrencephaly of the pinniped cerebral cortex has been noted in a study of fossil endocasts, in which pinniped ancestors appear to have cerebral cortices that are less gyrencephalic than extant pinnipeds (Edinger, 1975). In contrast to the cetaceans and pinnipeds, the manatee (and, although not quantified, other representatives of the Sirenia; Pirlot and Kamiya, 1985) has a well-expressed lissencephaly of the cerebral cortex (Reep and O'shea, 1990). The telencephalon of the manatee is dominated by enlarged lateral ventricles (Welker, 1990), and this increase in size of the lateral ventricles may be one factor in preventing the cortex of this species from forming sulci and gyri during development, perhaps as a result of the internal tension/pressure created by the larger lateral ventricle preventing significant cortical folding (Van Essen, 1997). Whatever the specific reasons for the evolution of extreme gyrencephaly or lissencephaly, all three lineages of marine aquatic mammals show distinct differences in the formation of gyri and sulci compared with terrestrial mammals. This would indicate that evolutionary pressures, of a nature quite different from that experienced by terrestrial mammals, are affecting brain structure in marine mammals (see, e.g., Manger, 2006). That these potential selection pressures specifically target the telencephalon in these secondarily aquatic species is also of interest and argues against popularized theories of marine mammal brain evolution (Marino et al., 2007, 2008).

Why are some mammals more gyrencephalic than others?

Although several theories regarding the formation of gyri and the extent of gyrification have been put forward, no consensus on this point has been reached. The current study, examining quantitatively the extent of gyrencephaly from seven mammalian orders, may help to elucidate this problem to some extent. In a previous study (Pillay and Manger, 2007), it was noted that, among the species examined (rodents, carnivores, primates, and artiodactyls), the artiodactyls were the most gyrencephalic. The increased relative gyrencephaly in the artiodactyls compared with other mammals was related to the thinner cerebral cortex of these mammals (Pillay and Manger, 2007). With a thinner cerebral cortex, it would be easier for the cortex to buckle mechanically, and the "gyral window" required for the cortical afferents and efferents (Prothero and Sundsten, 1984) would be smaller because of the lower number of neurons found in the artiodactyls compared with carnivores and primates (115,000–175,000 per mm^3 in artiodactyls; Chow et al., 1950). The current study underscores this point by demonstrating that the cetaceans, known to have a very thin cerebral cortex for their brain size (which ranges between 1.3 and 1.75 mm; Ridgway and Brownson, 1984) and a low neuronal density (13,000–65,000 per mm^3; Manger, 2006), have the most extensive gyrencephaly of all mammals studied to date. This thin cetacean cortex will allow the cortex to mechanically buckle more readily during development, and having a low neuronal number that will

decrease the extent of cortical afferents and efferents will reduce the size of any required gyral windows. It is thus perhaps not surprising that cetaceans are the most gyrencephalic mammals. Unfortunately, data on the relative thickness and neuronal density of the cerebral cortex and of pinnipeds in relation to other carnivores are not currently available, but one might expect that the pinnipeds will show a thinner cerebral cortex with a lower neuronal density than carnivores with a similarly sized brain. If this expectation is met, it may explain why the pinnipeds are highly gyrencephalic carnivores. The manatee, on the other hand, shows a highly lissencephalic cortex. The manatees are known to have quite a thick cerebral cortex of 2.5–3 mm (Reep et al., 1989; Marshall and Reep, 1995) and a cortical neuronal density of approximately 30,000 neurons/mm^3 (Prof. Chet Sherwood, personal communication). The cortical thickness of the manatee is about double that of the cetaceans but is not dissimilar to that seen in other large brained mammals (such as humans and elephants), although the neuronal density is similar to that seen in cetaceans. Thus, for the manatee, it is unlikely that cortical thickness and cortical connections are preventing the formation of gyri, and sulci and perhaps other features, such as the large lateral ventricles, must be invoked to explain manatee lissencephaly. In this sense, the lissencephaly of the manatee cerebral cortex does not fit well with the gyral window hypothesis.

The current study has highlighted, through analyzing a broad range of mammalian species, the importance of both cortical thickness and neuronal number in determining the extent of cortical folding. As discussed above, the more highly folded cortices appear to be those that are thinner and have lower cortical neuronal densities. Although these observations lend some support to the gyral window hypothesis of cortical folding, several other concepts related to cortical folding have not yet been discussed. As mentioned above, the order-specific placement of sulci and gyri has been observed across a range of mammalian species (detailed to the greatest extent by Welker, 1990, among many others). If we examine the most highly gyrencephalic mammals, the cetaceans and artiodactyls, there are certain major sulci and gyri that are consistent within and between species throughout these orders (Welker, 1990); however, the smaller sulci and gyri do appear to be extremely variable between individuals, especially those with larger brains. Other concepts postulated to correlate with cortical folding, or to be causal factors inducing cortical folding, are the developmental buckling of the cortex related to bonded layers showing varying cellular density (Richman et al., 1975), friction of the cortical sheet with subcortical structures (His, 1874) and the subplate during development (Armstrong et al., 1995), or tension associated with connectivity in the developing and adult brain (Van Essen, 1997; Hilgetag and Barbas, 2005). As mentioned above, the tension-based hypothesis appears to provide an explanation for the lack of gyrification in the manatee cerebral cortex; however, to associate any of these mechanisms with the extent of gyrencephaly across the range of mammalian species examined is not a straightforward task. For example, the tension-based hypothesis would posit a higher number of cortical areas in the cetaceans than other mammals, but cetaceans have far fewer cortical areas than are found in the less gyrencephalic humans (Manger, 2006). It would also posit a greater extent of both short- and long-range connectivity in cetaceans, something that again appears to be absent in cetaceans (Manger et al., 2010). Unfortunately, developmental studies detailing the relationship between the cortex with the subplate or other subcortical structures are not available across a range of species, so the extent to which these developmental mechanisms may play a role in cortical folding cannot be assessed.

At present, it appears that a multifactorial approach to cortical gyrification, with phylogenetically sensitive approaches, may provide the most complete answer. This approach would require a great deal more empirical data on features such as neuronal densities, cortical thicknesses, quantitative connectivity patterns, developmental timing schedules, and morphological integration aspects of cortical and subcortical structures, in order to tease out which of the factors that have been shown to be related to cortical gyrification are indeed causal and which are merely correlative. The current study does provide some support for the gyral window hypothesis of gyrencephaly in mammals and demonstrates that a combination of thinner cerebral cortex and lower neuronal density leads to higher gyrencephalic indices; however, it is clear that, to understand how, why, and to what extent the cerebral cortex (and other neural structures such as cerebellar cortex) has fissures and folds, a great deal of further information is required.

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